

TOP 10 ML MODELS FOR CYBER THREAT INTELLIGENCE

Model Name

Primary Use Case

Why it's "Best" for your Project

1

XGBoost

Multi-class Classification

Fastest inference time; perfect for real-time dashboard updates.

2

Isolation Forest

Anomaly Detection

Identifies "unknown" threats without needing labeled attack logs.

3

LSTM (RNN)

Temporal Trend Analysis

Captures sequential patterns; essential for "Trend Detection" module.

4

Random Forest

Malware Detection

Highly robust; prevents overfitting on small or imbalanced datasets.

5

BERT

Log Parsing/NLP

Transforms raw text logs into structured data for better visualization.

6

Autoencoders

Reconstructive Anomaly

Flags subtle deviations by failing to "reconstruct" malicious traffic.

7

CNN

Traffic Pattern Recognition

Converts network flow into "images" to detect visual attack patterns.

8

K-Means

Behavior Clustering

Groups similar suspicious IPs to identify botnets on your map.

9

SVM

Perimeter Defense

High precision for binary classification (Safe vs. Malicious).

10

Naive Bayes

Phishing/Spam Filter

Extremely lightweight; ideal for high-speed preprocessing in Module 1.

How These Models Work in Your Project

XGBoost: The Decision Engine

XGBoost (Extreme Gradient Boosting) works by building a series of decision trees where each new tree corrects the errors of the previous one. In your project, it takes the structured logs from Module 1 and predicts the specific attack type (e.g., DDoS, SQLi, or Brute Force) with over 99% accuracy. Because it is optimized for speed, your dashboard can refresh almost instantly.

Isolation Forest: The Outlier Detector

Instead of learning what an attack looks like, this model learns how to "isolate" data. Since attacks are rare and different from normal traffic, they require fewer "splits" to be isolated in a tree structure. This provides the Anomaly Score used in your "Trend and Anomaly Detection" outcome.

LSTM: The Time-Series Forecaster

LSTMs are designed for data with a time dimension. It remembers previous states of network traffic to determine if the current sequence is malicious. This allows your dashboard to show spikes and seasonality—identifying if an attack is a one-time event or part of a slow, multi-day pattern.

The Absolute Best Model: XGBoost

After research, XGBoost is the best model for a visualization-heavy project like yours for three specific reasons:

Low Latency: In 2026 benchmarks, XGBoost is roughly 14% faster than Random Forest in inference time. For an interactive dashboard, speed is everything.

Explainability (SHAP/Feature Importance): It doesn't just give a "Yes/No." It can tell you which feature (e.g., bytes_out or dst_port) led to the risk score. You can display this directly in your Vulnerability Prioritization charts.

Handling Imbalance: Cybersecurity data is famously imbalanced (mostly normal traffic). XGBoost handles this better than deep learning models without needing massive computational power.

